Embedded Systems Lab

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Task 2

Traffic Light System

- Prerequisite
- Personal realization of task 1 uploaded to git
- Extend the (car) traffic light system of the first sprint by a pedestrian light
- Extend a button for the pedestrian. If pressed, the pedestrian lights switches as fast as possible to green.
- Guarantee that the pedestrian lights signal is red if the traffic light signal is green
- If the pedestrian lights signal is green the traffic light is red.
- At least the green traffic light is required to stay for 10 time units in green and 5 time units in yellow.
- First, model your solution
- Extend the class diagram
- Add a sequence diagram for showing the interaction with a pedestrian pressing a button including the switching of the lights
- Update the state machine model accordingly
- Map the state machine model according to the **switch case pattern** to Arduino code
- Extend the tinkercad simulation by a pedestrian button
- Green and red light for the pedestrian light
- Upload your personal solution to <gitrep>/lab/<yourname>
- Deadline: one day before next class (groups A,B,C,D 20 Oct,E,F 23 Oct)